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Inventor(s)

STIGALL; Jeremy et al.

FLEXIBLE SUPPORT MEMBER FOR INTRALUMINAL IMAGING DEVICE AND ASSOCIATED DEVICES, SYSTEMS, AND METHODS

Abstract

Intraluminal ultrasound imaging devices, methods, and systems are provided. Some embodiments of the present disclosure include intraluminal devices configured to navigate and maneuver tortuous vasculature of a patient. For example, an intravascular imaging device can include a flexible support member at a distal portion of the imaging device. The flexible support member may support a plurality of ultrasound transducers and a plurality of control circuits. The flexible support member includes a flexible joint portion between a transducer portion and a controller portion. Introducing a flexible joint portion between the transducer portion and the controller portion can provide for greater flexibility of the imaging device and enable the device to navigate tortuous regions of a patient's anatomy.

Inventors: STIGALL; Jeremy (CARLSBAD, CA), SONG; Jun (SAN DIEGO, CA)

Applicant: PHILIPS IMAGE GUIDED THERAPY CORPORATION (SAN DIEGO, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 16/967,531, filed Aug. 5, 2020, now U.S. Pat. No. 12,303,327, which is the national stage entry of International Application No. PCT/EP2019/052308, filed Jan. 31, 2019, which claims the benefit of and priority to U.S. Provisional Application No. 62/628,340, filed Feb. 9, 2018, each of which is incorporated by reference herein.

TECHNICAL FIELD

[0002] The present disclosure relates generally to intraluminal imaging, such as intravascular ultrasound (IVUS) imaging, and, in particular, to the distal structure of an intraluminal imaging device. For example, the distal structure can include a support member and a flex circuit that are arranged to facilitate efficient assembly and operation of the intraluminal imaging device.

BACKGROUND

[0003] Intravascular ultrasound (IVUS) imaging is widely used in interventional cardiology as a diagnostic tool for assessing a diseased vessel, such as an artery, within the human body to determine the need for treatment, to guide the intervention, and/or to assess its effectiveness. An IVUS device including one or more ultrasound transducers is passed into the vessel and guided to the area to be imaged. The transducers emit ultrasonic energy in the form of ultrasonic waves in order to create an image of the vessel of interest. The ultrasonic waves are partially reflected by discontinuities arising from tissue structures (such as the various layers of the vessel wall), red blood cells, and other features of interest. The reflected waves are received by the transducer and passed along to an IVUS imaging system. The imaging system processes the received ultrasound echoes to produce a cross-sectional image of the vessel where the device is placed.

[0004] Solid-state (also known as synthetic-aperture) IVUS catheters are one of the two types of IVUS devices commonly used today, the other type being the rotational IVUS catheter. Solid-state IVUS catheters carry a scanner assembly that includes an array of ultrasound transducers distributed around its circumference along with one or more integrated circuit controller chips mounted adjacent to the transducer array. The controllers select individual transducer elements (or groups of elements) for transmitting an ultrasound pulse and for receiving the ultrasound echo signal. By stepping through a sequence of transmit-receive pairs, the solid-state IVUS system can synthesize the effect of a mechanically scanned ultrasound transducer but without moving parts (hence the solid-state designation). Since there is no rotating mechanical element, the transducer array can be placed in direct contact with the blood and vessel tissue with minimal risk of vessel trauma. Furthermore, because there is no rotating element, the electrical interface is simplified. The solid-state scanner can be wired directly to the imaging system with a simple electrical cable and a standard detachable electrical connector, rather than the complex rotating electrical interface required for a rotational IVUS device.

[0005] Manufacturing an intravascular imaging device that can efficiently traverse physiology within the human body is challenging. In that regard, components at the distal portion of the imaging device cause an area of high rigidity in the intravascular device, which increases the

likelihood of kinking as the intravascular device is steered through tortuous vasculature. In some instances, the area of high rigidity can limit the ability of the imaging device to navigate and maneuver through particularly tortuous regions of the vasculature.

SUMMARY

[0006] The present disclosure advantageously describes intravascular imaging devices configured to navigate and maneuver tortuous vasculature of a patient. For example, an intravascular imaging device can include a flexible support member at a distal portion of the imaging device. The flexible support member includes a flexible joint portion between a transducer portion and a controller portion. Introducing a flexible joint portion between the transducer portion and the controller portion can provide for greater flexibility of the imaging device and enable the device to navigate tortuous regions of a patient's anatomy.

[0007] In some aspects, an intraluminal ultrasound imaging device includes a plurality of ultrasound transducer elements configured to obtain imaging data, a plurality of control circuits in communication with the plurality of ultrasound transducer elements, a plurality of conductive traces extending between the plurality of ultrasound transducer elements and the plurality of control circuits, a flexible elongate body configured to be positioned within a body lumen of a patient, and a support member coupled to a distal portion of the flexible elongate body. The support member can include a transducer portion adjacent to the plurality of ultrasound transducer elements, a control portion adjacent to the plurality of control circuits, and a flex joint portion comprising a flexible element disposed between the transducer portion and the control portion. The flex joint portion is adjacent to the plurality of conductive traces, and the joint portion is more flexible than the transducer portion and the control portion, in some embodiments.

[0008] In some embodiments, the support member is generally cylindrical. The support member may comprise stainless steel. In some embodiments, the support member can include a polymer and a conductive agent. The flex joint portion can include a plurality of cuts. In other embodiments, the polymer may include an acoustic dampening material. In still other embodiments, the flex joint portion comprises a coil coupled to the transducer portion and the control portion. The flexible substrate may be positioned around the support member, in some aspects, and the plurality of conductive traces may be coupled to the flexible substrate. The support member can include an elongate body and one or more flanges coupled to the elongate body, wherein an outer diameter of the one or more flanges is greater than an outer diameter of the elongate body. The device may further include an acoustic backing material disposed between the elongate body and the plurality of ultrasound transducer elements. The flexible substrate can include one or more slot openings configured to facilitate flexing of the flexible substrate. The flexible substrate can also comprise one or more grounding regions in communication with the support member.

[0009] According to some embodiments, the plurality of ultrasound transducer elements comprises an ultrasonic transducer array disposed in an annular configuration around the transducer portion of the support member. In other embodiments, the device includes a shrink tubing positioned around and configured to hermetically seal at least one of: the plurality of ultrasound transducer elements; the plurality of control circuits; or the plurality of communication lines. The support member can further include a first flange at a proximal portion of the transducer portion and a second flange at a distal portion of the transducer portion, wherein the first flange and second flange are configured to space the plurality of transducer elements from the support member. In some aspects, the support member includes a third flange at a proximal portion of the control portion and a fourth flange at a distal portion of the control portion. The third flange and fourth flange can be configured to support the plurality of control circuits.

[0010] According to other aspects of the present disclosure, an intraluminal ultrasound imaging system is provided that includes an intraluminal imaging device and a computing device. The intraluminal imaging device includes a flexible elongate body configured to be positioned within a body lumen of a patient, and an imaging assembly disposed at a distal portion of the flexible

elongate body and configured to obtain imaging data associated with the body lumen. The imaging assembly includes a support member around which a plurality of transducer elements, a plurality of control circuits, and a plurality of communication lines are positioned. The support member can include a transducer portion adjacent to the plurality of transducer elements, a control portion adjacent to the plurality of control circuits, and a flex joint portion. The flex joint portion can comprise a flexible element disposed between the transducer portion and the control portion. The flex joint portion can be adjacent to the plurality of communication lines, and can be more flexible than the transducer portion and the control portion. In some aspects, the computing device is in communication with the intraluminal imaging device and configured to output, to a display, an image of the body lumen based on the obtained imaging data.

[0011] In some embodiments, the intraluminal imaging device includes a catheter. In other embodiments, the intraluminal imaging device comprises an intravascular ultrasound (IVUS) device. Additional aspects, features, and advantages of the present disclosure will become apparent from the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Illustrative embodiments of the present disclosure will be described with reference to the accompanying drawings, of which:

[0013] FIG. 1 is a diagrammatic schematic view of an intraluminal imaging system, according to aspects of the present disclosure.

[0014] FIG. 2 is a diagrammatic top view of a flex circuit in a flat configuration, according to aspects of the present disclosure.

[0015] FIG. 3 is a diagrammatic side view of a flex circuit in a rolled configuration around a support member, according to aspects of the present disclosure.

[0016] FIG. 4 is a diagrammatic cross-sectional side view of a distal portion of an intravascular device, according to aspects of the present disclosure.

[0017] FIG. 5 is a diagrammatic top view of an intravascular device in a tortuous portion of a blood vessel, according to aspects of the present disclosure.

[0018] FIG. 6 is a diagrammatic top view of a flex circuit in a flat configuration, according to aspects of the present disclosure.

[0019] FIG. 7 is a diagrammatic top view of a conductive trace region of a flex circuit, according to aspects of the present disclosure.

[0020] FIG. 8 is a diagrammatic top view of a conductive trace region of a flex circuit, according to aspects of the present disclosure.

[0021] FIG. 9 is a diagrammatic top view of a conductive trace region of a flex circuit, according to aspects of the present disclosure.

[0022] FIG. 10 is a diagrammatic top view of a portion of a flex circuit in a flat configuration, according to aspects of the present disclosure.

[0023] FIG. 11 is a diagrammatic top view of a portion of a flex circuit in a flat configuration, according to aspects of the present disclosure.

[0024] FIG. 12 is a diagrammatic side view of a support member, according to aspects of the present disclosure.

[0025] FIG. 13 is a diagrammatic side view of a support member and a flex circuit in a rolled configuration, according to aspects of the present disclosure.

[0026] FIG. 14 is a diagrammatic side view of a support member, according to aspects of the present disclosure.

[0027] FIG. 15 is a diagrammatic side view of a support member, according to aspects of the

present disclosure.

[0028] FIG. **16** is a diagrammatic side view of a polymeric support member and a flex circuit in a rolled configuration, according to aspects of the present disclosure.

[0029] FIG. **17** is a diagrammatic cross-sectional view of a distal portion of a support member and a flex circuit in a rolled configuration, according to some aspects of the present disclosure.

[0030] FIG. **18** is a diagrammatic top view of an intravascular imaging device in a tortuous portion of a blood vessel, according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0031] For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It is nevertheless understood that no limitation to the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, and methods, and any further application of the principles of the present disclosure are fully contemplated and included within the present disclosure as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately.

[0032] FIG. **1** is a diagrammatic schematic view of an intraluminal imaging system **100**, according to aspects of the present disclosure. For example the system **100** can be an intravascular ultrasound (IVUS) imaging system. The imaging system **100** may include an intraluminal ultrasound imaging device **102** (such as a solid-state or phased array IVUS device), a patient interface module (PIM) **104**, a processing system or console **106**, and a monitor **108**. The device **102** can be a catheter guide wire, or guide catheter.

[0033] At a high level, the IVUS device **102** emits ultrasonic energy from a transducer array **124** included in a scanner assembly **110** mounted near a distal end of the catheter device. For the purposes of this disclosure, the scanner assembly **110** may also be referred to as a flex circuit or an imaging assembly, in some instances. The ultrasonic energy is reflected by tissue structures in the medium, such as a vessel **120**, surrounding the scanner assembly **110**, and the ultrasound echo signals are received by the transducer array **124**. The PIM **104** transfers the received echo signals to the console or computer **106** where the ultrasound image (including the flow information) is reconstructed and displayed on the monitor **108**. The console or computer **106** can include a processor and a memory. The computer or computing device **106** can be operable to facilitate the features of the IVUS imaging system **100** described herein. For example, the processor can execute computer readable instructions stored on a non-transitory tangible computer readable medium.

[0034] The PIM **104** facilitates communication of signals between the IVUS console **106** and the scanner assembly **110** included in the IVUS device **102**. This communication includes the steps of: (1) providing commands to control logic dies **206A**, **206B**, illustrated in FIG. **2**, included in the scanner assembly **110** to select the particular transducer array element(s) to be used for transmit and receive, (2) providing the transmit trigger signals to the control logic dies **206A**, **206B** included in the scanner assembly **110** to activate the transmitter circuitry to generate an electrical pulse to excite the selected transducer array element(s), and/or (3) accepting amplified echo signals received from the selected transducer array element(s) via amplifiers included on the control logic die(s) **206** of the scanner assembly **110**. In some embodiments, the PIM **104** performs preliminary processing of the echo data prior to relaying the data to the console **106**. In examples of such embodiments, the PIM **104** performs amplification, filtering, and/or aggregating of the data. In an embodiment, the PIM **104** also supplies high-and low-voltage DC power to support operation of the device **102** including circuitry within the scanner assembly **110**.

[0035] The console **106** receives the echo data from the scanner assembly **110** by way of the PIM

104 and processes the data to reconstruct an image of the tissue structures in the medium surrounding the scanner assembly **110**. The console **106** outputs image data such that an image of the vessel **120**, such as a cross-sectional image of the vessel **120**, is displayed on the monitor **108**. Vessel **120** may represent fluid filled or surrounded structures, both natural and man-made. The vessel **120** may be within a body of a patient. The vessel **120** may be referenced as a body lumen of the patient. The body lumen or vessel **120** may be a blood vessel, such as an artery or a vein of a patient's vascular system, including cardiac vasculature, peripheral vasculature, neural vasculature, renal vasculature, and/or or any other suitable lumen inside the body. For example, the device **102** may be used to examine any number of anatomical locations and tissue types, including without limitation, organs including the liver, heart, kidneys, gall bladder, pancreas, lungs; ducts; intestines; nervous system structures including the brain, Dural sac, spinal cord and peripheral nerves; the urinary tract; as well as valves within the blood, chambers or other parts of the heart, and/or other systems of the body. In addition to natural structures, the device **102** may be may be used to examine man-made structures such as, but without limitation, heart valves, stents, shunts, filters and other devices.

[0036] In some embodiments, the IVUS device includes some features similar to traditional solid-state IVUS catheters, such as the EagleEye® catheter available from Volcano Corporation and those disclosed in U.S. Pat. No. 7,846,101 hereby incorporated by reference in its entirety. For example, the IVUS device **102** includes the scanner assembly **110** near a distal end of the device **102** and a transmission line bundle **112** extending along the longitudinal body of the device **102**. The transmission line bundle or cable **112** can include a plurality of conductors, including one, two, three, four, five, six, seven, or more conductors **218** (FIG. 2). It is understood that any suitable gauge wire can be used for the conductors **218**. In an embodiment, the cable **112** can include a four-conductor transmission line arrangement with, e.g., 41 AWG gauge wires. In an embodiment, the cable **112** can include a seven-conductor transmission line arrangement utilizing, e.g., 44 AWG gauge wires. In some embodiments, 43 AWG gauge wires can be used.

[0037] The transmission line bundle **112** terminates in a PIM connector **114** at a proximal end of the device **102**. The PIM connector **114** electrically couples the transmission line bundle **112** to the PIM **104** and physically couples the IVUS device **102** to the PIM **104**. In an embodiment, the IVUS device **102** further includes a guide wire exit port **116**. Accordingly, in some instances the IVUS device is a rapid-exchange catheter. The guide wire exit port **116** allows a guide wire **118** to be inserted towards the distal end in order to direct the device **102** through the vessel **120**.

[0038] FIG. 2 is a top view of a portion of an ultrasound scanner assembly, or flex circuit **214**, according to an embodiment of the present disclosure. The flex circuit **214** includes a transducer array **124** formed in a transducer region **204** and transducer control logic dies **206** (including dies **206A** and **206B**) formed in a control region **208**, with a transition region **210** disposed therebetween. For the purposes of this disclosure, the transducer control logic dies **206** may also be referred to as control circuits. The transducer control logic dies **206** and the transducers **212** are mounted on a flexible substrate or film whereon various electrical components can be formed and/or attached. FIG. 3 illustrates a rolled configuration of the flex circuit **214**. The transducer array **124** is a non-limiting example of a medical sensor element and/or a medical sensor element array. The transducer region **204** is disposed adjacent a distal portion **220** of the flex circuit **214**. The control region **208** is disposed adjacent the proximal portion **222** of the flex circuit **214**. The transition region **210** is disposed between the control region **208** and the transducer region **204**. Dimensions of the transducer region **204**, the control region **208**, and the transition region **210** (e.g., lengths **225**, **227**, **229**) can vary in different embodiments. In some embodiments, the lengths **225**, **227**, **229** can be substantially similar or a length **227** of the transition region **210** can be greater than lengths **225**, **229** of the transducer region and controller region, respectively. While the embodiments of FIGS. 2 and 3 are described as including a flex circuit, it is understood that the transducers and/or controllers may be arranged in other configurations, including those omitting a

flex circuit.

[0039] The transducer array **124** may include any number and type of ultrasound transducers **212**, or ultrasound transducer elements, although for clarity only a limited number of ultrasound transducers are illustrated in FIG. 2. In an embodiment, the transducer array **124** includes **64** individual ultrasound transducers **212**. In a further embodiment, the transducer array **124** includes **32** ultrasound transducers **212**. Other numbers are both contemplated and provided for. With respect to the types of transducers, in an embodiment, the ultrasound transducers **212** are piezoelectric micromachined ultrasound transducers (PMUTs) fabricated on a microelectromechanical system (MEMS) substrate using a polymer piezoelectric material, for example as disclosed in U.S. Pat. No. 6,641,540, which is hereby incorporated by reference in its entirety. In alternate embodiments, the transducer array includes piezoelectric zirconate transducers (PZT) transducers such as bulk PZT transducers, capacitive micromachined ultrasound transducers (cMUTs), single crystal piezoelectric materials, other suitable ultrasound transmitters and receivers, and/or combinations thereof.

[0040] The flex circuit **214** may include various transducer control logic, which in the illustrated embodiment is divided into discrete control logic dies **206**. In various examples, the control logic of the flex circuit **214** performs: decoding control signals sent by the PIM **104** across the cable **112**, driving one or more transducers **212** to emit an ultrasonic signal, selecting one or more transducers **212** to receive a reflected echo of the ultrasonic signal, amplifying a signal representing the received echo, and/or transmitting the signal to the PIM across the cable **112**. In the illustrated embodiment, the flex circuit **214**, having **64** ultrasound transducers **212** divides the control logic across nine control logic dies **206**, of which five are shown in FIG. 2. Designs incorporating other numbers of control logic dies **206** including 8, 9, 16, 17 and more are utilized in other embodiments. In general, the control logic dies **206** are characterized by the number of transducers they are capable of driving, and exemplary control logic dies **206** drive 4, 8, and/or 16 transducers.

[0041] The control logic dies are not necessarily homogenous. In some embodiments, a single controller is designated a master control logic die **206A** and contains the communication interface for the cable **112**. Accordingly, the master control circuit may include control logic that decodes control signals received over the cable **112**, transmits control responses over the cable **112**, amplifies echo signals, and/or transmits the echo signals over the cable **112**. The remaining controllers are slave controllers **206B**. The slave controllers **206B** may include control logic that drives a transducer **212** to emit an ultrasonic signal and selects a transducer **212** to receive an echo. In the depicted embodiment, the master controller **206A** does not directly control any transducers **212**. In other embodiments, the master controller **206A** drives the same number of transducers **212** as the slave controllers **206B** or drives a reduced set of transducers **212** as compared to the slave controllers **206B**. In an exemplary embodiment, a single master controller **206A** and eight slave controllers **206B** are provided with eight transducers assigned to each slave controller **206B**.

[0042] The flex circuit **214**, on which the transducer control logic dies **206** and the transducers **212** are mounted, provides structural support and interconnects for electrical coupling. The flex circuit **214** may be constructed to include a film layer of a flexible polyimide material such as KAPTON™ (trademark of DuPont). Other suitable materials include polyester films, polyimide films, polyethylene naphthalate films, or polyetherimide films, other flexible printed semiconductor substrates as well as products such as Upilex® (registered trademark of Ube Industries) and TEFLON® (registered trademark of E.I. du Pont). In the flat configuration illustrated in FIG. 2, the flex circuit **214** has a generally rectangular shape. As shown and described herein, the flex circuit **214** is configured to be wrapped around a support member **230** (FIG. 3) to form a cylindrical toroid in some instances. Therefore, the thickness of the film layer of the flex circuit **214** is generally related to the degree of curvature in the final assembled flex circuit **214**. In some embodiments, the film layer is between 5 μm and 100 μm, with some particular embodiments being between 12.7 μm and 25.1 μm.

[0043] To electrically interconnect the control logic dies **206** and the transducers **212**, in an embodiment, the flex circuit **214** further includes conductive traces **216** formed on the film layer that carry signals between the control logic dies **206** and the transducers **212**. In some instances, the conductive traces **216** can be referred to as communication lines. In particular, the conductive traces **216** providing communication between the control logic dies **206** and the transducers **212** extend along the flex circuit **214** within the transition region **210**. In some instances, the conductive traces **216** can also facilitate electrical communication between the master controller **206A** and the slave controllers **206B**. The conductive traces **216** can also provide a set of conductive pads that contact the conductors **218** of cable **112** when the conductors **218** of the cable **112** are mechanically and electrically coupled to the flex circuit **214**. Suitable materials for the conductive traces **216** include copper, gold, aluminum, silver, tantalum, nickel, and tin, and may be deposited on the flex circuit **214** by processes such as sputtering, plating, and etching. In an embodiment, the flex circuit **214** includes a chromium adhesion layer. The width and thickness of the conductive traces **216** are selected to provide proper conductivity and resilience when the flex circuit **214** is rolled. In that regard, an exemplary range for the thickness of a conductive trace **216** and/or conductive pad is between 10-50 μm . For example, in an embodiment, 20 μm conductive traces **216** are separated by 20 μm of space. The width of a conductive trace **216** on the flex circuit **214** may be further determined by the width of the conductor **218** to be coupled to the trace/pad.

[0044] The flex circuit **214** can include a conductor interface **240**, or interface region, in some embodiments. The conductor interface **240** can be a location of the flex circuit **214** where the conductors **218** of the cable **114** are coupled to the flex circuit **214**. For example, the bare conductors of the cable **114** are electrically coupled to the flex circuit **214** at the conductor interface **240**. The conductor interface **240** can be a tab extending from the main body of flex circuit **214**. In that regard, the main body of the flex circuit **214** can refer collectively to the transducer region **204**, controller region **208**, and the transition region **210**. In the illustrated embodiment, the conductor interface **240** extends from the proximal portion **222** of the flex circuit **214**. In other embodiments, the conductor interface **240** is positioned at other parts of the flex circuit **214**, such as the distal portion **220**, or the flex circuit **214** omits the conductor interface **240**. A value of a dimension of the tab or conductor interface **240**, such as a width **224**, can be less than the value of a dimension of the main body of the flex circuit **214**, such as a width **226**. In some embodiments, the substrate forming the conductor interface **240** is made of the same material(s) and/or is similarly flexible as the flex circuit **214**. In other embodiments, the conductor interface **240** is made of different materials and/or is comparatively more rigid than the flex circuit **214**. For example, the conductor interface **240** can be made of a plastic, thermoplastic, polymer, hard polymer, etc., including polyoxymethylene (e.g., DELRIN®), polyether ether ketone (PEEK), nylon, and/or other suitable materials. As described in greater detail herein, the support member **230**, the flex circuit **214**, the conductor interface **240** and/or the conductor(s) **218** can be variously configured to facilitate efficient manufacturing and operation of the flex circuit **214**.

[0045] In some instances, the flex circuit **214** is transitioned from a flat configuration (FIG. 2) to a rolled or more cylindrical configuration (FIGS. 3 and 4). For example, in some embodiments, techniques are utilized as disclosed in one or more of U.S. Pat. No. 6,776,763, titled "ULTRASONIC TRANSDUCER ARRAY AND METHOD OF MANUFACTURING THE SAME" and U.S. Pat. No. 7,226,417, titled "HIGH RESOLUTION INTRAVASCULAR ULTRASOUND TRANSDUCER ASSEMBLY HAVING A FLEXIBLE SUBSTRATE," each of which is hereby incorporated by reference in its entirety.

[0046] As shown in FIGS. 3 and 4, the flex circuit **214** is positioned around the support member **230** in the rolled configuration. FIG. 3 is a diagrammatic side view of an intravascular imaging device **102** with the flex circuit **214** in a rolled configuration around a support member **230**, according to aspects of the present disclosure. FIG. 4 is a diagrammatic cross-sectional side view of a distal portion of the intravascular device **102**, including the flex circuit **214** and the support

member **230**, according to aspects of the present disclosure.

[0047] The support member **230** can be referenced as a unibody in some instances. The support member **230** can be composed of a metallic material, such as stainless steel, or non-metallic material, such as a plastic or polymer as described in U.S. Provisional Application No. 61/985,220, "Pre-Doped Solid Substrate for Intravascular Devices," filed Apr. 28, 2014, the entirety of which is hereby incorporated by reference herein. The support member **230** can be ferrule having a distal portion **262** and a proximal portion **264**. The support member **230** can define a lumen **236** extending longitudinally therethrough. The lumen **236** is in communication with the exit port **116** and is sized and shaped to receive the guide wire **118** (FIG. 1). The support member **230** can be manufactured accordingly to any suitable process. For example, the support member **230** can be machined, such as by removing material from a blank to shape the support member **230**, or molded, such as by an injection molding process. In some embodiments, the support member **230** may be integrally formed as a unitary structure, while in other embodiments the support member **230** may be formed of different components, such as a ferrule and flanges **242**, **244**, that are fixedly coupled to one another.

[0048] Flanges **242**, **244** that extend outward from the support member **230** are provided at the distal and proximal portions **262**, **264**, respectively, of the support member **230**. The flanges **242**, **244** support the distal and proximal portions of the flex circuit **214**. In that regard, portions of the flex circuit **214**, such as the transducer region **204**, can be spaced from a central body portion of the support member **230** extending between the flanges **242**, **244**. The flanges **242**, **244** can have the same outer diameter or different outer diameters. For example, the distal flange **242** can have a larger or smaller outer diameter than the proximal flange **244**. To improve acoustic performance, a cavity between the flex circuit **214** and the surface of the support member **230** can be filled with a backing material **246**. The backing material **246** can be introduced between the flex circuit **214** and the support member **230** via passageways **235** in the flanges **242**, **244**. In some embodiments, suction can be applied via the passageways **235** of one of the flanges **242**, **244**, while the backing material **246** is fed between the flex circuit **214** and the support member **230** via the passageways **235** of the other of the flanges **242**, **244**. In some embodiments, the backing material can be a liquid. The backing material can be cured to allow it to solidify and set. In various embodiments, the support member **230** includes more than two flanges **242**, **244**, only one of the flanges **242**, **244**, or neither of the flanges. In that regard the support member **230** can have an increased diameter distal portion **262** and/or increased diameter proximal portion **264** that is sized and shaped to elevate and support the distal and/or proximal portions of the flex circuit **214**.

[0049] The support member **230** can be substantially cylindrical in some embodiments. Other shapes of the support member **230** are also contemplated including geometrical, non-geometrical, symmetrical, non-symmetrical, cross-sectional profiles. Different portions the support member **230** can be variously shaped in other embodiments. For example, the proximal portion **264** can have a larger outer diameter than the outer diameters of the distal portion **262** or a central portion extending between the distal and proximal portions **262**, **264**. In some embodiments, an inside diameter of the support member **230** (e.g., the diameter of the lumen **236**) can correspondingly increase or decrease as the outer diameter changes. In other embodiments, the inside diameter of the support member **230** remains the same despite variations in the outer diameter. In other embodiments, the inside diameter of the support member **230** can change while the outer diameter remains the same.

[0050] In some embodiments, the transducer elements **212**, array **124**, and/or the controllers **206** can be positioned in an annular configuration, such as a circular configuration or in a polygon configuration, around a longitudinal axis LA of the imaging device **102** (FIG. 1). For example, a cross-sectional profile of the imaging assembly **110** at the transducer elements **212**, array **124**, and/or the controllers **206** can be a circle or a polygon. Any suitable annular polygon shape can be implemented, such as a based on the number of controllers/transducers, flexibility of the

controllers/transducers, etc., including a pentagon, hexagon, heptagon, octagon, nonagon, decagon, etc. The cross-section shape of the imaging device **102** in the regions **208**, **210**, **204** may be different. For example, the cross-sectional shape of the region **208** can be a nonagon, the cross-sectional shape of the region **210** can be approximately circular, and the cross-sectional shape of the region **204** can be polygonal (with the number of sides of the polygon equaling the number of transducer elements **212** in the array **124**).

[0051] A proximal inner member **256** and a proximal outer member **254** are coupled to the proximal portion **264** of the support member **230**. The proximal inner member **256** and/or the proximal outer member **254** can be a flexible elongate body that extends from proximal portion of the intravascular device **102**, such as the proximal connector **114**, to the flex circuit **214**. For example, the proximal inner member **256** can be received within the proximal portion **264**. The proximal outer member **254** abuts and is in contact with the flex circuit **214**. A distal member **252** is coupled to the distal portion **262** of the support member **230**. The distal member **252** can be a flexible component that defines a distal most portion of the intravascular device **102**. For example, the distal member **252** is positioned around the distal portion **262**. The distal member **252** can abut and be in contact with the flex circuit **214** and the flange **242**. The distal member **252** can be the distal-most component of the intravascular device **102**.

[0052] One or more adhesives can be disposed between various components at the distal portion of the intravascular device **102**. For example, one or more of the flex circuit **214**, the support member **230**, the distal member **252**, the proximal inner member **256**, and/or the proximal outer member **254** can be coupled to one another via an adhesive.

[0053] FIG. 5 illustrates an embodiment of a distal portion of an intravascular imaging device **102** inside a vessel **10** of a patient. The vessel **10** comprises a bend that can be partially characterized by an inside diameter **16**, and a circle **15** having a radius **18**. In that regard the bend of the blood vessel **10** can be described as following a portion of the circumference of the circle **15**. The imaging device **102** can comprise a plurality of the components described above with respect to FIGS. 3 and 4, including a support member and a flex circuit. In some embodiments, the imaging device **102** has a so-called “stiff length” **70** partially characterized by the geometry of the support member and the flex circuit. The stiff length **70** may determine the types of vasculature through which the imaging device **102** can traverse.

[0054] For example, in embodiments comprising a rigid support member having a length **L**, the stiff length **70** of the distal portion of the imaging device **102** may be the length **L**. The stiff length **70** may also be characterized, in part, by a width of the distal portion of the imaging device **102**. As the distal portion of the imaging device **102** navigates to the bend in the blood vessel **10**, the shape, size, and configuration of the distal portion of the imaging device **102**, including the stiff length **70**, can limit the regions of the patient's vasculature through which the imaging device **102** can travel. The bend in the vessel **10** of FIG. 5 may be the most tortuous path through which the imaging device **102** can traverse. In other words, the imaging device **102** of FIG. 5 may not be able to pass through vessels having a sharper bend than the vessel **10** of FIG. 5 and/or vessels having an inside diameter smaller than the inside diameter **16** of the vessel **10** of FIG. 5. Attempting to force an imaging device having a relatively large stiff length through tortuous vasculature could result in damage to the vessel **10** tissue or structure.

[0055] FIG. 6 illustrates exemplary embodiment of a flex circuit **314** or flexible substrate in a flat configuration. At least a portion of the flex circuit **314**, such as a transition region **310** can be shaped and configured to improve flexibility of the intravascular device. As described below with respect to FIG. 13, a support member **330** around which the flex circuit **314** is positioned can be shaped to compliment the orientation of the flex circuit **314** around the support member **330**. The flex circuit **314** and the support member **330** can be shaped to advantageously increase the flexibility of intravascular device and decrease the likelihood of kinking while the intravascular device is navigated through a patient's vasculature. The flex circuit **314** can be positioned at a distal

portion of an intravascular device (e.g., device **102**).

[0056] The flex circuit **314** can be similar in some respects to the flex circuit **214** of FIG. 2. The flex circuit **314** includes a transducer region **304** having a plurality of transducers **312** at a distal portion **320** and a controller region **308** having plurality of control circuits **306** at a proximal portion **322**. A transition region **310** having a plurality of conductive traces **316** extending between the controller and transducer regions **308**, **304** facilitates communication between the plurality of transducers **312** and the plurality of control circuits **306**. The flex circuit **314** can comprise a flexible substrate on which the conductive traces **316** are formed. Similar to the flex circuit **214** of FIG. 2, the transition region **310** can have a width of any suitable value, including between approximately 0.2 mm and 10 mm. The transducer region **304** and/or the controller region **308** can have widths between approximately 1 mm and 10 mm, for example. Another dimension of the flex circuit **314**, such as a length, can have a larger value than a corresponding dimension, such as lengths of the transducer region **304** and the controller region **308**, respectively. The length of the transition region can be any suitable value, including between approximately 0.1 mm and 12 mm. The length of the transducer region **304** can be between approximately 0.5 mm and 6 mm for example. The length of the controller region **308** can be between approximately 0.5 mm and 8 mm for example.

[0057] The flex circuit **314** can comprise a flexible substrate or film. The transition region **310** of the flex circuit **314** can comprise one or more relief cuts **301**, **303**, **305**, slots, or openings, configured to provide enhanced flexibility of the flex circuit **314** to be wrapped, rolled, or otherwise disposed around a support member. The relief cuts **301**, **303**, **305** extend longitudinally along the length of the flexible substrate **314** within the transition region **310**. The cuts **301**, **303**, **305** may extend parallel to one another. Referring to the embodiment of FIG. 6, the flex circuit **314** comprises three relief cuts **301**, **303**, **305** configured to increase flexibility of the flex circuit **314**. In some embodiments, the flex circuit **314** can include more or fewer relief cuts than the embodiment illustrated in FIG. 6. For example, the flexible substrate **314** can include zero, one, two, three, four, five, six, or more cuts. In some embodiments, the flex circuit **314** comprises no relief cuts. In some embodiments, the flex circuit **314** can comprise other means of enhancing flexibility, such as one or more perforations, creases, and/or areas of decreased thickness of the flexible substrate.

[0058] The transition region **310** is depicted in a trapezoidal fashion for illustrative purposes only, and does not necessarily depict the layout of the flex circuit **314**. For example, the transition region **310** may be rectangular, and the relief cuts **301**, **303**, **305**, may comprise linear cuts. In other embodiments, the transition region **310** may comprise a parallelogram shape, a concave shape, or any other suitable shape.

[0059] The flex circuit **314** includes grounding regions **315**, **317**, including a distal grounding region **315** and a proximal grounding region **317**. The distal grounding region **315** can be disposed near a distal end **319** of the flex circuit **314**, distal of the plurality of transducers **312**. In other embodiments, the distal grounding region **315** can be disposed proximal to the transducers **312**. The proximal grounding region **317** can be disposed near a proximal portion **324** of the flex circuit **314**. As further explained below, the grounding regions **315**, **317** can be configured to abut or contact one or more conductive regions, such as flanges, of a support member to facilitate grounding of one or more components of the flex circuit **314** (e.g., transducers **312** and controllers **306**). In some embodiments, the flex circuit **314** can include more or fewer grounding regions than the embodiment illustrated in FIG. 6. For example, in some embodiments, a flex circuit **314** can include a single grounding region configured to facilitate electrical grounding for the flex circuit **314**. In other embodiments, the flex circuit **314** may have no grounding regions. In yet other embodiments, the flex circuit **314** can include 3, 4, or more grounding regions.

[0060] The flex circuit **314** comprises an interface region **350** at a proximal portion **324** of the flex circuit. The interface region **350** comprises a plurality of components configured to interface and/or couple to other components of an intravascular imaging device and/or imaging system. Interface

region **350** comprises a plurality of capacitors **352** which can be configured to modify imaging data received by the plurality of transducers **312**. In some embodiments, the capacitors **352** can be configured to reduce noise included in the imaging data obtained by the plurality of transducers **312**. Interface region **350** also includes a plurality of connector regions **354** and testing regions **356**. The connector regions **354** can comprise conductive regions or pads configured to provide an electrical connection between the flex circuit **314** and other components of an imaging device and/or imaging system. For example, in some embodiments, the connector regions **354** comprise a plurality of soldering pads. Testing regions **356** can be configured similarly to the connector regions **354**, in some aspects. In some embodiments, the testing regions **356** may be in communication with the connector regions **354**. The testing regions **356** can be configured to provide a communication interface with the components of the flex circuit **314** during manufacturing, assembly, and/or testing of the flex circuit. The testing regions **356** can be removed from the interface region **350** during or after assembly, in some cases. Although disposed at the proximal portion **324** of the flex circuit **314**, the interface region **350** can be disposed at other regions of the flex circuit, such as near the distal end **319**, or near the transition region **310**.

[0061] FIGS. **7-9** respectively depict trace patterns of conductive traces of the controller region **308**, the transition region **310**, and the transducer region **304**. Referring to FIG. **7**, the conductive traces **316** of the controller region **308** may be arranged in an angled or slanted configuration with respect to an axis of the flex circuit **314**. Referring to FIG. **8**, the conductive traces **316** in the transition region **310** may be arranged in a linear or straight configuration with respect to an axis of the flex circuit **314**. Referring to FIG. **9**, the conductive traces **316** in the transducer region **304** may be arranged in a counter angle, or “zig-zag” configuration. Some of the patterns or arrangements shown in FIGS. **7-9** may increase the robustness of the flex circuit **314** in operation. For example, the angled configuration of FIG. **7** and the “zig-zag” configuration of FIG. **9** may protect the conductive traces **316** during frequent movements that apply strain, stress, and/or compression to the conductive traces **316** near the controller region **308** and/or the transducer region **304**. The configurations of FIGS. **7** and **9** may allow for the flex circuit **314** to be rolled and installed on a support member, and to navigate tortuous regions of the patient's vasculature while mitigating a risk of kinking or fracturing one or more of the conductive traces **316**.

[0062] This disclosure contemplates other configurations and patterns of conductive traces **316** than those shown in FIGS. **7-9**. For example, in some embodiments, the conductive traces **316** at the transducer region **304** can have an angled configuration similar to that of FIG. **7**. Moreover, in some embodiments, various regions of the flex circuit **314** can include conductive traces **316** in a zig-zag arrangement. In some embodiments, some or all of the regions of the flex circuit **314** can include an angled configuration similar to that of FIG. **7**, and/or a straight configuration such as that of FIG. **8**. Although not explicitly described, this disclosure contemplates other combinations of conductive trace patterns suitable to communicatively couple the plurality of transducers **312** to the plurality of control circuits **306**.

[0063] FIGS. **10** and **11** illustrate a distal portion of a flex circuit **314**, including a transducer region **304**, according to some embodiments. In particular, FIGS. **10** and **11** can illustrate the correspondingly labeled region of FIG. **6** at the distal region of the flex circuit **314**. In some aspects, the flex circuits **314** shown in FIGS. **10** and **11** can be the same flex circuit **314** shown in FIG. **6**. In FIG. **10**, the conductive traces **316** are formed on the flex circuit **314** in a zig-zag formation at a region **311**, and in a linear configuration at a region **312**. The conductive traces **316** can continue to other regions of the flex circuit **314**, such as the controller region **308**, to communicatively couple the transducers **312** to the control circuits **306**. In FIG. **11**, the conductive traces **316** are formed in an angled configuration in the region **311** and a linear configuration in the region **312**. As described above with respect to FIGS. **7-9**, the zig-zag and/or angled configurations of the conductive traces **316** can increase the durability and longevity of the conductive traces **316** during use of the imaging device. As the flex circuit **314** of the imaging device bends, flexes, and

strains while passing through the tortuous vasculature of the patient, the zig-zag and/or angled configurations of the conductive traces **316** may mitigate risk of fracture and/or kinking of the conductive traces **316** to maintain communication of the transducers **312** with the control circuits **306**. While FIGS. **10** and **11** illustrate the distal portion of the flex circuit **314**, the disclosure can similarly apply to other portions of the flex circuit **314**, such as the conductive traces leading to or extending from the controllers **306**.

[0064] FIG. **12** illustrates a side view of the support member **330**. The support member **330** can be composed of a metallic material, such as stainless steel, or non-metallic material, such as a plastic or polymer as described in U.S. Provisional Application No. 61/985,220, "Pre-Doped Solid Substrate for Intravascular Devices," filed Apr. 28, 2014, the entirety of which is hereby incorporated by reference herein. The support member **330** has a controller portion **338**, a transducer portion **334**, and a flex joint portion **336** extending between the controller and transducer portions **338**, **334**. The support member **330** may be variously sized and shaped in different embodiment. For example, while the illustrated embodiment shows a substantially cylindrical support member **330**, the support member **330** may comprise other shapes, such as a rectangular prism, ellipsoid, and/or other suitable shape. The support member **330** may be sized and shaped to allow greater flexibility of the intravascular device. For example, the support member **330** may compliment the size and shape the flex circuit **314**.

[0065] The controller and transducer portions **338**, **334** can have an outer profile that is substantially cylindrically-shaped. The flex joint portion **336** can be a bridge extending between the controller and transducer portions **338**, **334**. In the illustrated embodiment, the flex joint portion **336** can be shaped in a cylindrical manner and extend in a linear manner between the controller and transducer portions **338**, **334**. For example, outer surfaces of the controller portion **338**, the flex joint portion **336**, and the transducer portion **334** may form a continuous surface. In other embodiments, the flex joint portion **336** may extend between the controller and transducer portions **338**, **334** such that outer surface of the support member **330** is discontinuous. Other configurations of the flex joint portion **336** are also contemplated.

[0066] One or more dimensions of the various portions of the support member **330** can correspond to one or more corresponding dimensions of the regions of the flex circuit **314**. In some instances, a circumference, width, and/or length of the various portions of the support member **330** can be based on cut, unrolled, and/or flattened state of the flex circuit **314**. Corresponding dimensions between the support member **330** and the flex circuit **314** advantageously facilitates support of the various regions **304**, **308**, **310** of the flex circuit **314**.

[0067] The controller and transducer portions **338**, **334** of the support member **330** can be variously sized and shaped to support the flex circuit when the flex circuit is positioned around the support member. The dimensions of the support member **330** can be selected such that the intravascular device **102** has a diameter between approximately 2 Fr and approximately 10 Fr, for example. In some embodiments, the controller and transducer portions **338**, **334** of the support member **330** are similarly sized and shaped. In other embodiments, the controller portion **338** can be larger or smaller than the transducer portion **334**. In some embodiments, a length of the controller portion **338** can be between approximately 3 mm and 7 mm, a length of the transducer portion **334** can be between approximately 0.5 mm to 2 mm, and a length of the flex joint portion can be between approximately 2 mm and 10 mm.

[0068] The controller, transducer, and flex joint portions **338**, **334**, **336** can be integrally formed. In some embodiments, the support member **330** can be shaped by removing material from a cylindrical blank. For example, the blank can be machined using a tool to cut, or remove material at cuts **335**. Forming the cuts **335** advantageously increases flexibility of the support member **330** by reducing the amount of material contributing to the stiffness of the support member **330**. In the illustrated embodiment, the cuts **335** can be removed from the flex joint portion **336**. The cuts **335** can extend radially inwards from an outer surface of the support member **330**. A lumen **360** can

extend longitudinally through the support member 330. The lumen 360 can include a proximal opening 362 at a proximal end of the support member 330, and a distal opening 364 at a distal end of the support member 330. The lumen 360 can be sized and shaped to allow a guide wire to pass therethrough.

[0069] The cuts 335 are shown in an alternating pattern. For example, the cuts 335 can be formed by cutting into the support member towards the lumen 360 from a top and bottom side of the support member 330 at a first plurality of points along the support member 330, and cutting radially inward into the support member 330 towards the lumen 360 from a front and back side of the support member 330 at a second plurality of points along the support member 330, wherein the first and second plurality of points are arranged in an alternating fashion. In some embodiments, the support member 330 is molded, such as with an injection molding process. In such instances, the support member 330 including the controller, transducer, and flex joint portions 338, 334, 336, are formed without removing the material indicated by cuts 335. The controller, transducer, and flex joint portions 338, 334, 336 can be integrally formed.

[0070] The support member 330 of FIG. 12 comprises a plurality of flanges 341, 342, 343, 344. The flanges 341, 342, 343, 344 may comprise washers or raised portions of the support member 330 at various points along the support member 330. In some embodiments, the flanges 341, 342, 343, 344 can be formed by securing a washer or flange to the cylindrical body 332 of the support member 330. In other embodiments, the flanges 341, 342, 343, 344 can be integrally formed with the other portions of the support member 330 by, for example, removing material from a cylindrical blank. A flange 341 is shown positioned at a proximal portion of the support member 330, near a proximal end of the controller portion 338, while a flange 342 is disposed near a distal end of the controller portion 338. In some aspects, the flanges 341, 342 can define the controller portion 338. The position of the flange 342 may also be described as the proximal end of the flex joint portion 336. A flange 343 is positioned near a distal end of the flex joint portion 336. In some aspects, the flanges 342, 343 can define the flex joint portion 336. The position of the flange 343 may also be described as the proximal end of the transducer portion 334. A flange 344 is positioned near a distal end of the transducer portion 334. In some aspects, the flange 343 and the flange 344 can define the transducer portion 334. The flanges 341, 342, 343, 344 are configured to support a flex circuit, such as the flex circuit 314 of FIG. 6, in a rolled or wrapped configuration. In some embodiments, the flanges 341, 342, 343, 344 can support the flex circuit 314 away from the cylindrical body 332 of the support member 330. The flanges 341, 342, 343, 344 can each have a diameter, such as diameter 349. In some embodiments, the flanges 341, 342, 343, 344 have the same diameter 349. In other embodiments, one or more of the flanges 341, 342, 343, 344 can have different diameters. The diameter of the flanges 341, 342, 343, 344 can be larger than a diameter of the body 332 of the support member 330.

[0071] One or more of the flanges 341, 342, 343, 344 can be electrically conductive to facilitate grounding. Each of the flanges 341, 342, 343, 344 can be placed to define the portions of the support member 330 to provide support and various components of the flex circuit 314. For example, the flanges 341, 342, which can define the controller portion 338, can provide support to the controller region 308 of the flex circuit 314 to protect and/or provide structure support to the control circuits 306. Similarly, the flanges 343, 344 can define the transducer portion 334 and can protect and/or provide structural support to the transducer region 304 of the flex circuit 314 to facilitate proper operation of the plurality of transducers 312. The flanges 341, 342, 343, 344 can also provide a space between the flex circuit and the cylindrical body 332 of the support member 330 to provide room for the components of the flex circuit 314. As described below, the flex circuit 314 may be wrapped or rolled around the support member 330 such that one or more components of the flex circuit 314 project radially inward toward the cylindrical body 332 of the support member 330, such that a smooth side of the flex circuit 314 is positioned on the outside of the intravascular device 300.

[0072] FIG. 13 is a diagrammatic side view of a distal portion of an ultrasonic imaging device 300. For example, an imaging assembly, such as the flex circuit 314 positioned around the support member 330, can be disposed at the distal portion of a flexible elongate member (e.g., a catheter). The controller, transducer, and flex joint portions 338, 334, 336 of the support member 330 support the flex circuit 314 while advantageously minimizing the amount of material of the support member 330, and increasing flexibility and maneuverability of the imaging device 300. The transducer region 304 and the controller region 308 can have a cylindrical or cylindrical toroid configuration when rolled around the support member 330. The lumen 360, sized and shaped to receive a guide wire, extends longitudinally through the support member 330. The transducer region 304 of the flex circuit 314 is positioned around transducer portion 334 of the support member 330 such that the transducers 312 are adjacent to the transducer portion 334. The controller region 308 of the flex circuit 314 is positioned around the controller portion 338 of the support member 330, such that the control circuits 306 are adjacent to the controller portion 338. The dimensions of the device 300 can be selected such that the device 300 has a diameter 349 between approximately 2 Fr and approximately 10 Fr, for example. In some embodiments, the diameter 349 of the device 300 may be substantially the same as the diameter 345 of one or more of the flanges 341, 342, 343, 344.

[0073] The transition region 310 of the flex circuit 314 is aligned with the flex joint portion 336 of the support member 330. The support member 330 can be shaped such that that flex joint portion 336 supports the transition region 310. In that regard, the transition region 310 can be adjacent to the flex joint portion 336 while the flex circuit 314 is positioned around the support member 330. The flex circuit 314 can be supported by the plurality of flanges 341, 342, 343, 344 with the components of the flex circuit projecting inward toward the cylindrical body 332, such that the outer surface of the imaging device 300 comprises a smooth surface.

[0074] In other embodiments, the transition region 310 of the flex circuit 314 extends between the controller and transducer regions 308, 304 in a linear, curved, spiral, and/or other suitable manner. Spiral wrapping of at least a portion of the flex circuit 314 is described in U.S. Provisional App. No. 62/315,395, titled "Single Unibody Designs", and filed Mar. 30, 2016, the entirety of which is hereby incorporated by reference herein. The dimensions of the flex circuit 314 and/or the support member 330 may vary in different embodiments. For example, the proximal and/or distal portions may be sized to respectively accommodate control circuits 306 and transducers 312. As described below with respect to FIG. 17, in some instances, the flex circuit 314 may be radially spaced from the support member 330, and an acoustic dampening material, or acoustic backing material 346, can be disposed between in a space between the flex circuit 314 and the support member 330 to facilitate operation of the transducers.

[0075] The flexible substrate 314 can be configured to hermetically seal one or more of the components of the substrate 314 (e.g., transducers 312, control circuits 306) in some embodiments. For example, the flexible substrate may be rolled or wrapped around the flanges 341, 342, 343, 344, so as to create hermetically sealed spaces between the substrate 314 and the support member 330. In some embodiments, the imaging device 300 may further comprise a flexible tubing configured to surround the flex circuit 314 once the flex circuit 314 has been rolled onto the support member 330. The flexible tubing may comprise a shrinkable tubing, in some instances. The shrinkable tubing may shrink in response to heat. The flexible tubing may comprise a low durometer, or soft, material. The flexible tubing may be configured to seal and/or protect one or more components of the flex circuit 314, such as the transducers 312 and/or the control circuits 306.

[0076] Generally, the imaging device 300 can be a catheter, a guide catheter, or a guide wire. The imaging device 300 includes a flexible elongate member, or elongate body (e.g., outer member 254 and/or inner member 256). The imaging assembly (e.g., flex circuit 314 and/or support member 330) can be disposed at the distal portion of the flexible elongate member. As used herein,

“elongate member” or “flexible elongate member” includes at least any thin, long, flexible structure structurally arranged (e.g., sized and/or shaped) to be positioned within a lumen of the anatomy. For example, a distal portion of the flexible elongate member is positioned within a lumen, while a proximal portion of the flexible elongate member is positioned outside of the body of the patient. The flexible elongate member can include a longitudinal axis. In some instances, the longitudinal axis can be a central longitudinal axis of the flexible elongate member. In some embodiments, the flexible elongate member can include one or more polymer/plastic layers formed of various grades of nylon, Pebax, polymer composites, polyimides, and/or Teflon. In some embodiments, the flexible elongate member can comprise stainless steel. In some embodiments, the flexible elongate member can include one or more layers of braided metallic and/or polymer strands. The braided layer(s) can be tightly or loosely braided in any suitable configuration, including any suitable per inch count (pic). In some embodiments, the flexible elongate member can include one or more metallic and/or polymer coils. All or a portion of the flexible elongate member may have any suitable geometric cross-sectional profile (e.g., circular, oval, rectangular, square, elliptical, etc.) or non-geometric cross-sectional profile. For example, the flexible elongate member can have a generally cylindrical profile with a circular cross-sectional profile that defines an outer diameter of the flexible elongate member. For example, the outer diameter of the flexible elongate member can be any suitable value for positioning within the anatomy, including between approximately 1 Fr and approximately 15 Fr, including values such as 3.5 Fr, 5 Fr, 7 Fr, 8.2 Fr, 9 Fr, and/or other suitable values both larger and smaller. Diameters **345** and **349** can have similar values in some instances.

[0077] FIG. **14** illustrates a diagrammatic side view of a support member **430**, according to some embodiments of the present disclosure. The support member **430** can be similar to the support members described herein, including the support members **230** and **330**, in some aspects. The support member **430** includes a flex joint portion **436** positioned between controller and transducer portions **438**, **434**. Similar to the flex joint portion **336** shown in FIG. **12**, the flex joint portion **436** of FIG. **14** advantageously enables the support member **430** to bend at the flex joint portion **436**, which may increase maneuverability of the intravascular device within vasculature.

[0078] The support member **430** of FIG. **14** may comprise fewer flanges **443**, **444** than the support member **330** shown in FIG. **12**. The support member **430** can comprise a flange **443** and a flange **444** at or near a transducer portion **434** of the support member **430**, and a grounding area **441** at or near a controller portion **438** of the support member **430**. The **443**, **444** can be configured to provide structural support to a corresponding transducer region of a flex circuit, such as the transducer region **304** of the flex circuit **314** of FIG. **6**. The support member **430** can also comprise a plurality of cuts **435** configured to increase flexibility of the support member **430** in the flex joint portion **436**.

[0079] The grounding area **441** may be described as a “solder bump” in some embodiments. The grounding area **441** may be configured to electrically couple to a corresponding grounding region of a flex circuit to facilitate grounding of the flex circuit via the support member **430**. The grounding area **441** of the support member **430** can comprise an area of raised material, such as a metal, configured to facilitate a secure electrical contact with a corresponding grounding region of a flex circuit. In some embodiments, the grounding area **441** can comprise a solder agent used during assembly of an intravascular device to create a solder connection of the grounding area **441** with a corresponding grounding region of a flex circuit in response to heat. In the illustrated embodiment, the transducer portion **334** includes flanges **443**, **444** configured to support a corresponding transducer region of a flex circuit, such as the transducer region **304** of the flex circuit **314**. The flanges **443**, **444** are configured to support the transducers **312** and transducer region **304** away from the support member **430** (e.g., radially spaced), such that a space is provided wherein an acoustic backing material can be included to improve performance of the transducers **312**. In contrast to the transducer region **304**, the controller region **308** may not require and/or

benefit from a space between the controllers **306** and the support member **430**. Thus, the support member **430** does not include flanges at the controller portion **438**. The controllers **306** and/or controller region **308** of the flex circuit **314** can surround and abut the support member **430** at or near the controller portion **438**.

[0080] FIG. **15** is a diagrammatic side view of a support member **530** to be used in an intravascular imaging device. The support member **530** can be similar to the support members described herein, including the support members **230**, **330**, and **430**, in some aspects. The support member **530** can be formed of different materials having differing degrees of flexibility. The support member **530** can also be formed of separate components that are fixedly joined. For example, the support member **530** can include a controller portion **538**, a transducer portion **534**, and a flex joint portion **536**. The flex joint portion **536** of the support member **530** can include a coil **535** or spring disposed between and coupled to the controller portion **538** and the transducer portion **534**. The coil **535** can be fixedly coupled to the other portions of the support member **530** in any suitable manner, including a solder and/or adhesives. In some embodiments, the coil **535** may be formed with the other portions of the support member **530** from a solid body, or unibody (e.g., a cylindrical blank), by machining and/or removing material from the flex joint portion **536** to create the coil **535** or spring. The coil **535** may behave similarly to the cuts **335**, **435** of the support members of FIGS. **12** and **14** to provide increased flexibility and maneuverability to an imaging device.

[0081] The support member of FIG. **14** includes a plurality of flanges **541**, **542**, **543**, **544** configured to support a flex circuit. The flanges **541**, **542**, **543**, **544** of the support member **530** may be configured in a similar manner as the flanges of the embodiment of FIGS. **12** and **13**. Each of the controller portion **538**, the flex joint portion **536**, and the transducer portion **534** may comprise a length corresponding to a length of a controller region, a transition region, and a transducer region of a flex circuit, respectively. The corresponding lengths of the portions of the support member may be greater than, less than, or substantially equal to the corresponding lengths of the portions of the support members **330**, **430** shown in FIGS. **12** and **14**. In some embodiments, the length of one portion of the support member of FIG. **14** may be greater than the length a corresponding portion of the support member **330** of FIG. **12**, while a length of another portion of the support member **430** of FIG. **14** is less than a length of the corresponding portion of the support member **330** of FIG. **12**, and vice versa.

[0082] FIG. **16** is a diagrammatic side view of an intravascular imaging device **600**, including a support member **630** and a flex circuit **614**, according to some embodiments of the present disclosure. The support member **630** of FIG. **16** can comprise an elongate body **632** formed of a polymer. The polymer may be malleable or soft such that it can be deformed to receive one or more components of the flex circuit **614**. For example, as the flex circuit **614**, including a plurality of transducers **612** and a plurality of control circuits **606**, is wrapped around the malleable polymer elongate body **632**, the polymer may deform such that the control circuits **606** and transducers **612** are partially embedded into the polymer elongate body **632**, and such that the flex circuit **614** lies substantially flat and/or flush against an outer surface of the elongate body **632**. In some embodiments, the support member **630** can include a conductive agent in the polymer to facilitate grounding of the flex circuit **614**. In some embodiments, the polymer elongate body **632** can comprise an acoustic dampening agent configured to improve performance of the transducers **612**. For example, the acoustic dampening agent can impede the propagation of ultrasonic energy emitted by the transducers **612** inward (e.g., towards the support member **630**), such that the ultrasonic energy is directed outward (e.g., towards the anatomy in which the intraluminal imaging device **600** is positioned). As the malleable polymer of the support member **630** can be conductive to facilitate grounding, and can deform to create a substantially flush or smooth outer surface of the imaging device, the support member **630** does not include supporting flanges such as those described in FIGS. **12-15**.

[0083] The support member **630** can include grounding portions **641**, **644** configured to facilitate

grounding between the grounding regions of the flex circuit (e.g., **315**, **317**), and the support member **630**. In some embodiments, the grounding portions **641**, **644** can comprise soldering pads wherein the corresponding grounding regions of the flex circuit can be soldered.

[0084] In some embodiments, the malleable polymer may be configured to harden, or set, in response to heat. A baking process may harden the support member to secure the flex circuit **614** to the elongate body **632**. Once hardened, the support member **630** may still be flexible in one or more aspects to navigate through tortuous regions of the vasculature of a patient. Thus, although the baking or hardening process may set or harden the support member **630** in some aspects, the support member **630** may still be capable of bending or flexing at one or more areas of the support member **630**.

[0085] FIG. **17** is a diagrammatic cross-sectional view of the transducer region of the imaging device of FIG. **13** taken along line **17-17**. The support member **330** is wrapped with the flex circuit **314** such that the transducer region **304** comprising the plurality of transducers **312** is at least partially disposed within a space between the flange **343** and the flange **344**. The space formed between the flange **343** and the flange **344** of the support member **330** and the flex circuit **314** can be filled with an acoustic backing material **346**. The acoustic backing material **346** can be a liquid in some instances. Other spaces formed between the flex circuit **314** and the support member **330**, such as the space **348** between the flange **343** and the flange **342**, may not comprise an acoustic backing material. The grounding region at the distal portion of the flex circuit **314** can be in contact with the flange **344** to facilitate electrical grounding of the flex circuit **314**. The lumen **360** can be partially defined by the elongate body **332** of the support member **330**, and can be configured to receive a guide wire.

[0086] FIG. **18** is a diagrammatic side view of a distal portion of an intravascular imaging device **700** shown traversing a tortuous region of a blood vessel **710**. FIG. **17** may illustrate similar aspects as those depicted in FIG. **5**. The intravascular imaging device **700** may comprise a support member and flex circuit as illustrated in FIGS. **12** and **13**, wherein the support member comprises a flex joint portion **736** between a controller portion **738** and a transducer portion **734**. The controller portion **738** and transducer portion **734** can be relatively more rigid than the flex joint portion **736**. The controller portion may be characterized by a stiff length **770**, and the transducer portion may be characterized by a stiff length **780**. Each of the stiff lengths **770**, **780** may be less than the stiff length **170** of the device **102** shown in FIG. **5**. The flex joint portion **736** may be configured to bend or flex to navigate tortuous regions of the vasculature, such as the bend of the blood vessel **710**. Having the controller portion **738** and transducer portion **734** separated by the flex joint portion **736** may enable the intravascular imaging device **700** to navigate more tortuous regions of the vasculature than the embodiment illustrated in FIG. **5**. For example, the blood vessel **710** shown in FIG. **17** may comprise an inside diameter **716** which is smaller than the inside diameter **16** of the blood vessel **10** of FIG. **5**. Furthermore, the blood vessel **710** of FIG. **17** may have a sharper bend than the bend of the blood vessel **10** illustrated in FIG. **5**. For example, the bend of the blood vessel **710** of FIG. **17** may be partially characterized by a circle **715** having a radius **718**, which is less than the radius **18** of the circle **15** of FIG. **5**. In other words, the device **102** of FIG. **5** may be unable to safely traverse the region of the blood vessel **710** illustrated in FIG. **17**. By contrast, the device **700** of FIG. **17** may be capable of safely traversing the region of the blood vessel **710**, which may be due, in part, to the implementation of the flex joint portion **736** between the controller portion **738** and transducer portion **734**.

[0087] Persons skilled in the art will recognize that the apparatus, systems, and methods described above can be modified in various ways. Accordingly, persons of ordinary skill in the art will appreciate that the embodiments encompassed by the present disclosure are not limited to the particular exemplary embodiments described above. In that regard, although illustrative embodiments have been shown and described, a wide range of modification, change, and substitution is contemplated in the foregoing disclosure. It is understood that such variations may

be made to the foregoing without departing from the scope of the present disclosure.
[0088] Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

Claims

1. An apparatus, comprising: an intravascular ultrasound (IVUS) catheter configured to be positioned within a blood vessel of a patient and comprising a flexible substrate in a rolled configuration, wherein the flexible substrate comprises: a plurality of transducer elements configured to obtain IVUS data; a plurality of control circuits in communication with the plurality of transducer elements; and a plurality of grounding regions configured to provide electrical ground for at least one of the plurality of transducer elements or the plurality of control circuits, wherein the plurality of grounding regions comprises a first proximal grounding region and a distal grounding region that are spaced from one another along a length dimension of the flexible substrate, wherein the first proximal grounding region and the distal grounding region are oriented to extend in a width dimension of the flexible substrate.
2. The apparatus of claim 1, wherein the first proximal grounding region and the distal grounding region each comprise a long axis and a short axis, wherein the long axis is oriented to extend in the width dimension.
3. The apparatus of claim 2, wherein a first transducer element and a second transducer elements are spaced from one another in the width dimension by a portion of the flexible substrate between the first transducer element and the second transducer element, wherein the long axis of the distal grounding region overlaps the first transducer element, the second transducer element, and the portion of the flexible substrate between the first transducer element and the second transducer element.
4. The apparatus of claim 2, wherein a first control circuit and a second control circuit are spaced from one another in the width dimension by a portion of the flexible substrate between the first control circuit and the second control circuit, wherein the long axis of the first proximal grounding region overlaps the first control circuit, the second control circuit, and the portion of the flexible substrate between the first control circuit and the second control circuit.
5. The apparatus of claim 1, wherein the flexible substrate comprises: a distal portion comprising the plurality of transducer elements and the distal grounding region; and a proximal portion comprising the plurality of control circuits and the first proximal grounding region.
6. The apparatus of claim 5, wherein the distal grounding region is distal of the plurality of transducer elements.
7. The apparatus of claim 5, wherein the proximal grounding region is proximal of the plurality of control circuits.
8. The apparatus of claim 1, wherein the plurality of grounding regions further comprises a second proximal grounding region distinct from the first proximal grounding region.
9. The apparatus of claim 8, wherein the first proximal grounding region and the second proximal grounding region are spaced from one another in the width dimension by a portion of the flexible substrate between the first proximal grounding region and the second proximal grounding region.
10. The apparatus of claim 9, wherein a long axis of the distal grounding region overlaps the first proximal grounding region, the second proximal grounding region, and the portion of the flexible substrate between the first proximal grounding region and the second proximal grounding region.
11. The apparatus of claim 9, wherein the IVUS catheter comprises a plurality of electrical wires, wherein the flexible substrate comprises an interface region extending in the length dimension from the portion of the flexible substrate between the first proximal grounding region and the second proximal grounding region, wherein the plurality of electrical wires is coupled to the flexible substrate at the interface region.

12. The apparatus of claim 1, wherein the IVUS catheter comprises a support member defining guidewire lumen, wherein the flexible substrate is in the rolled configuration around the support member, wherein the plurality of grounding regions contact the support member.
